Task 05 Excel - Track A: Excel Formulas and Visualizations

Course: Database Applications Development
Lesson: 05 - Excel Integration with SQL Data

Overview

In this task, you will work with the Excel files you exported from your SQL queries. You'll practice Excel skills that align with the Microsoft Office Specialist (MOS) Excel Associate certification.

Files you should have:

1. `team_performance.xlsx`
2. `top_scorers.xlsx`
3. `win_loss_records.xlsx`
4. `high_scoring_games.xlsx`

If you don't have these files, complete the SQL task first!

Skills Practiced (MOS Excel Associate Aligned)

- Create and modify formulas (SUM, AVERAGE, MAX, MIN, IF, COUNTIF)
- Format cells and ranges
- Create and format charts
- Sort and filter data
- Apply conditional formatting
- Use absolute and relative cell references

Part 1: team_performance.xlsx - Formulas and Calculations

Open: `team_performance.xlsx`

Task 1.1: Add Calculated Columns

Add these new columns with formulas:

Column: Total Points (Season)

- Formula location: Create new column after `avg_points`
- Formula: `=games_played * avg_points`
- Purpose: Calculate approximate total season points
- Format: Number, 0 decimals

Column: Win Percentage

- Formula location: Create new column after `losses`
- Formula: `=(wins / games_played) * 100`
- Purpose: Calculate win percentage
- Format: Number, 1 decimal place, add % symbol

Column: Performance Rating

- Formula location: Final column
- Formula: `=IF(win_pct > 60, "Excellent", IF(win_pct > 50, "Good", IF(win_pct > 40, "Average", "Below Average")))`
- Purpose: Categorize teams by performance
- Format: Text, center aligned

Task 1.2: Add Summary Statistics

At the bottom of your data, add these calculations:

Row: League Average

- Label in first column: "League Average"
- Calculate: `=AVERAGE(range)` for games, wins, avg_points, etc.
- Format: Bold, fill with light gray

Row: Best Team

- Label: "Best Performance"
- Use: `=MAX(range)` for relevant columns
- Format: Bold, fill with light green

Row: Worst Team

- Label: "Needs Improvement"
- Use: `=MIN(range)` for relevant columns
- Format: Bold, fill with light red

Task 1.3: Conditional Formatting

Apply conditional formatting:

1. **Wins column:**
    - Color scale: Red (low) → Yellow (mid) → Green (high)
2. **Avg Points column:**
    - Data bars: Blue bars showing relative values
3. **Performance Rating column:**
    - Highlight rules:
        - "Excellent" = Green fill
        - "Good" = Yellow fill

    - "Average" = Orange fill
    - "Below Average" = Red fill

Task 1.4: Create Charts

Chart 1: Top 10 Teams by Wins

- Type: Horizontal bar chart (bar chart rotated)
- Data: Top 10 teams sorted by wins
- X-axis: Team names
- Y-axis: Number of wins
- Title: "Top 10 Teams - Win Total (2021-22)"
- Position: Below your data table

Chart 2: Win % Distribution

- Type: Column chart
- Data: All teams, sorted by win percentage
- X-axis: Team names (or abbreviations if available)
- Y-axis: Win percentage
- Title: "NBA Team Win Percentage (2021-22)"
- Add: Horizontal line at 50% (=.500 mark)
- Position: New sheet or next to Chart 1

Part 2: top_scorers.xlsx - Statistical Analysis

Open: `top_scorers.xlsx`

Task 2.1: Add Statistical Formulas

Add these columns:

Column: Points Per Game (verify)

- Formula: `=total_points / games_played`
- Purpose: Double-check the ppg calculation
- Format: Number, 1 decimal
- Compare to existing PPG column (should match)

Column: Scoring Tier

- Formula: `=IF(ppg >= 25, "Elite", IF(ppg >= 20, "All-Star", IF(ppg >= 15, "Starter", "Role Player")))`
- Purpose: Categorize players by scoring output
- Format: Text, center

Column: Games Percentage

- Formula: `=(games_played / 82) * 100`
- Purpose: What % of season did they play?

- Format: Number, 0 decimals, % symbol
- Note: Full NBA season is 82 games

Task 2.2: Count and Summary Functions

Create a summary section:

Count by Tier:

- Use `=COUNTIF(tier_range, "Elite")` for each tier
- Show count of Elite, All-Star, Starter, Role Player
- Format as a small table off to the side

Top Performers:

- Highest PPG: `=MAX(ppg_range)` and use `=INDEX(MATCH(...))` to find player name
- Lowest PPG (in top 50): `=MIN(ppg_range)`
- Average PPG: `=AVERAGE(ppg_range)`

Task 2.3: Create Scoring Charts

Chart 1: Top 15 Scorers

- Type: Horizontal bar chart
- Data: Top 15 by PPG
- X-axis: Player names
- Y-axis: Points per game
- Title: "Top 15 Scorers - PPG (2021-22)"
- Color: Vary colors by tier if possible

Chart 2: Scoring Tier Distribution

- Type: Pie chart
- Data: Count of players in each tier
- Labels: Tier names with counts
- Title: "Player Distribution by Scoring Tier"
- Show percentages

Part 3: win_loss_records.xlsx - Comparative Analysis

Open: `win_loss_records.xlsx`

Task 3.1: Add Win/Loss Analysis

Add columns:

Column: Games Behind Leader

- Formula: `=MAX($wins_range) - wins`
- Purpose: How many wins behind the best team?
- Format: Number, 0 decimals

- Use: Absolute reference for MAX range

Column: Winning Streak Potential

- Formula: `=82 - games_played`
- Purpose: Games remaining (if season not complete)
- Format: Number, 0 decimals

Column: Win Pace (82-game projection)

- Formula: `=(wins / games_played) * 82`
- Purpose: Project wins over full season
- Format: Number, 1 decimal

Task 3.2: Playoff Cutoff Analysis

Create analysis section:

Calculate:

- Playoff cutoff (typically 8th place in each conference)
- Use `=LARGE(wins_range, 8)` to find 8th most wins
- Show how many teams are "In the Hunt" (within 5 games)
- Formula: `=COUNTIF(games_behind_range, "<=5")`

Task 3.3: Create Win/Loss Charts

Chart 1: Win-Loss Records

- Type: Stacked bar chart
- Data: All teams
- Series 1: Wins (green)
- Series 2: Losses (red)
- X-axis: Team names
- Title: "Win-Loss Records by Team"
- Sort by total wins

Chart 2: Win Percentage Scatter

- Type: Scatter plot
- X-axis: Games played
- Y-axis: Win percentage
- Title: "Win % by Games Played"
- Purpose: See consistency across teams

Part 4: high_scoring_games.xlsx - Game Analysis

Open: `high_scoring_games.xlsx`

Task 4.1: Game Analysis Formulas

Add columns:

Column: Scoring Category

- Formula: `=IF(points >= 140, "Historic", IF(points >= 130, "Exceptional", IF(points >= 120, "High", "Standard")))`
- Purpose: Categorize scoring levels
- Format: Text, center

Column: Win/Loss Result

- If not already included, show W or L from your data
- Format: Text, center
- Apply conditional formatting (W=Green, L=Red)

Column: Month

- Extract month from game_date
- Formula: `=TEXT(game_date, "mmmm")` or `=MONTH(game_date)`
- Purpose: See seasonal patterns
- Format: Text

Task 4.2: Summary Statistics

Calculate:

- Total high-scoring games: `=COUNTA(points_range)`
- Average points in these games: `=AVERAGE(points_range)`
- Highest score: `=MAX(points_range)`
- Teams with most 120+ games: `=COUNTIF(team_range, team_name)`

Create pivot summary:

- Count games by team
- Count games by month
- Win/loss split

Task 4.3: Create Trend Charts

Chart 1: High Scoring Games Over Time

- Type: Line chart or scatter
- X-axis: Game date
- Y-axis: Points scored
- Title: "High-Scoring Games Throughout Season"
- Add: Trendline to show if scoring increased

Chart 2: Teams with Most 120+ Games

- Type: Column chart
- Data: Count of games per team
- X-axis: Team names

- Y-axis: Number of 120+ point games
- Title: "Teams with Most High-Scoring Games"

Formatting Requirements

Professional presentation:

1. **Headers:**
    - Bold, larger font (14pt)
    - Fill color (blue or team color)
    - White text
    - Centered
2. **Data:**
    - Consistent number formatting
    - Aligned appropriately (text left, numbers right)
    - Borders around cells for clarity
3. **Charts:**
    - Clear, descriptive titles
    - Axis labels
    - Legend when needed
    - Positioned neatly
    - Consistent color scheme
4. **Overall:**
    - No empty awkward spaces
    - Professional color scheme
    - Readable fonts (11-12pt for data)
    - Print-ready (fits on pages appropriately)

Submission Checklist

Before submitting, verify:

- [ ] All 4 Excel files completed
- [ ] All required formulas added and working
- [ ] All charts created with proper formatting
- [ ] Conditional formatting applied
- [ ] Summary statistics calculated
- [ ] Professional formatting throughout
- [ ] Files saved with your updates
- [ ] All sheets named appropriately

Submit:

- All 4 completed Excel files
- Push to GitHub in your `databaseApplications` folder

Rubric

Category	Points	Criteria
Formulas	40	All required formulas working correctly
Charts	30	All charts created, properly formatted
Formatting	15	Professional presentation, conditional formatting
Analysis	10	Summary statistics, insights calculated
Completeness	5	All tasks finished, files submitted
Total	**100**	

Tips for Success

1. **Use cell references**, not hard-coded values
2. **Test formulas** with different data
3. **Format as you go** - don't wait until the end
4. **Check your math** - do results make sense?
5. **Chart readability** - can someone else understand it?
6. **Save frequently** - Excel can crash!
7. **Ask for help** if stuck - don't struggle silently

You're learning real-world analyst skills! 📊

Next Steps

After completing this task:

- You've practiced SQL aggregations
- You've exported data for analysis
- You've created Excel formulas and visualizations
- You're ready for database design (Lesson 6)!

This is the complete data analysis workflow! SQL for processing → Excel for presentation.